CHEMISTRY ONLY

28 ESTERS

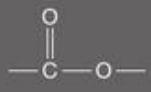
Alcohols, carboxylic acids and esters are organic compounds that contain oxygen as well as carbon and hydrogen. They are closely related. Carboxylic acids are made from oxidation of alcohols. In this chapter, you will learn how we can make esters by reacting alcohols with carboxylic acids. Esters are frequently used for perfumes and artificial flavours.



▲ Figure 28.1 Fats and oils contain the ester functional group.

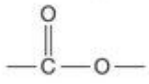
LEARNING OBJECTIVES

- Know that esters contain the functional group



- Know that ethyl ethanoate is the ester produced when ethanol and ethanoic acid react in the presence of an acid catalyst
- Understand how to write the structural and displayed formulae of ethyl ethanoate
- Understand how to write the structural and displayed formulae of an ester, given the name or formula of the alcohol and carboxylic acid from which it is formed and vice versa
- Know that esters are volatile compounds with distinctive smells and are used as food flavourings and in perfumes
- Practical: Prepare a sample of an ester such as ethyl ethanoate

Esters are organic compounds formed by the reaction of an alcohol with a carboxylic acid. They have the functional group



This can be written in condensed form as **–COO–**.

MAKING A SIMPLE ESTER: ETHYL ETHANOATE

Heating a mixture of ethanoic acid and ethanol with a few drops of concentrated sulfuric acid produces a liquid called ethyl ethanoate.

$$\underset{\text{ethanoic acid}}{CH_3COOH(l)} + \underset{\text{ethanol}}{CH_3CH_2OH(l)} \rightleftharpoons \underset{\text{ethyl ethanoate}}{CH_3COOCH_2CH_3(l)} + \underset{\text{water}}{H_2O(l)}$$

Starting materials: ethanol and ethanoic acid
Catalyst: concentrated sulfuric acid
Conditions: heat

DID YOU KNOW?

Ethyl ethanoate has quite a strong smell. It can be used as a solvent in nail polish and nail polish remover.

The concentrated sulfuric acid isn't written into the equation because it is a *catalyst* and it isn't consumed in the reaction.

The reaction is called **esterification**. It can also be described as a **condensation** reaction *because water is made when two molecules are joined together*. We will talk more about condensation reactions in the next chapter.

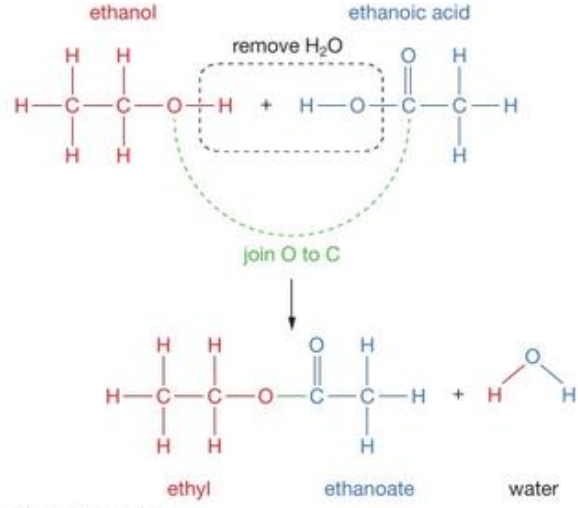
This reaction is *reversible* so to maximise the yield of ethyl ethanoate we often use pure ethanol and ethanoic acid (pure ethanoic acid is called *glacial ethanoic acid*). In Chapter 21 we discussed how the position of equilibrium always shifts to counteract any changes you make. If water is added to the reactants, the system would want to shift to the left to consume the water and therefore decrease the amount of ethyl ethanoate made. That is not what we want if we plan to make an ester!

DRAWING AND NAMING ESTERS

In order to work out the structure of the ester formed when an alcohol and a carboxylic acid react together, it is easier to start by drawing the alcohol and carboxylic acid so that their OH groups are next to each other. Now remove H_2O and join together what is left.

EXTENSION WORK

In an esterification reaction, an H from the alcohol and an OH from the acid are removed to form water. However, this distinction is not very important at International GCSE level. If you remove OH from the alcohol and H from the acid, you will obtain the same structure if you just join together what is left over.

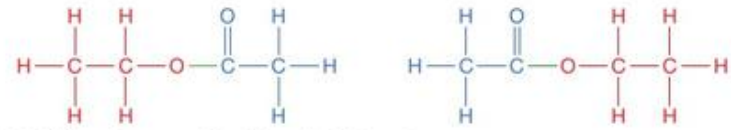


▲ Figure 28.2 Formation of an ester

An ester can be drawn either way round (see Figure 28.3) but the name is always written the same way round: ethyl ethanoate.

KEY POINT

When naming esters the part with the C=O is always the 'oate' part.



▲ Figure 28.3 Different ways of drawing ethyl ethanoate

Some other examples are shown in Table 28.1.

Table 28.1 Examples of esters

Alcohol \ Acid	methanoic acid	ethanoic acid	propanoic acid
methanol	methyl methanoate	methyl ethanoate	methyl propanoate
ethanol	ethyl methanoate	ethyl ethanoate	ethyl propanoate
propan-1-ol	propyl methanoate	propyl ethanoate	propyl propanoate

KEY POINT

The colours in Table 28.1 are used to show how the esters are made. The parts in red come from the alcohols and the parts in blue come from the carboxylic acids.

KEY POINT

It isn't actually important which C–O bond you break, as long as you add H to one side and OH to the other to give two OH groups you will get the same answer.

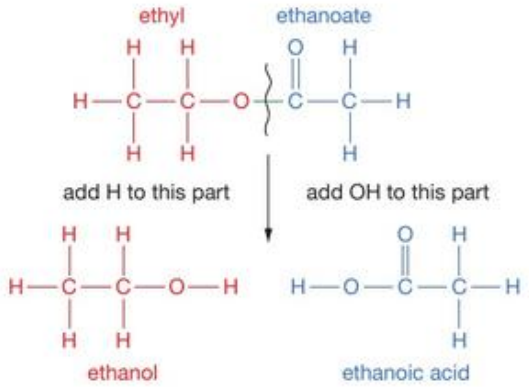
HINT

When working out the alcohol and carboxylic acid that an ester comes from, the '**yl**' bit tells you the alcohol and the '**oate**' bit tells you the carboxylic acid. So, prop**yl** butan**oate** is made from propanol and butanoic acid.

DID YOU KNOW?

Esters are often described as having a sweet, fruity smell.

You also need to be able to reverse this and work out which carboxylic acid and alcohol you would have to react together to produce a given ester. To do this, break apart the molecule between the C and O and add OH to one side and H to the other so that you have two OH groups.



▲ Figure 28.4 Splitting an ester

$$\underset{\text{ethyl ethanoate}}{CH_3COOCH_2CH_3(l)} + \underset{\text{water}}{H_2O(l)} \rightleftharpoons \underset{\text{ethanoic acid}}{CH_3COOH(l)} + \underset{\text{ethanol}}{CH_3CH_2OH(l)}$$

USES OF ESTERS

The small esters like ethyl ethanoate are commonly used in solvents, but most esters are used in food flavourings and perfumes. They are *volatile* liquids (*they turn to vapour easily*) with distinctive smells. The typical smell of bananas, raspberries, pears and other fruit is due in part to naturally occurring esters. Chemists create artificial flavourings and perfumes using mixtures of esters and other organic compounds.

ACTIVITY 1

▼ PRACTICAL: MAKING AND TESTING THE SMELL OF SOME ESTERS IN THE LAB

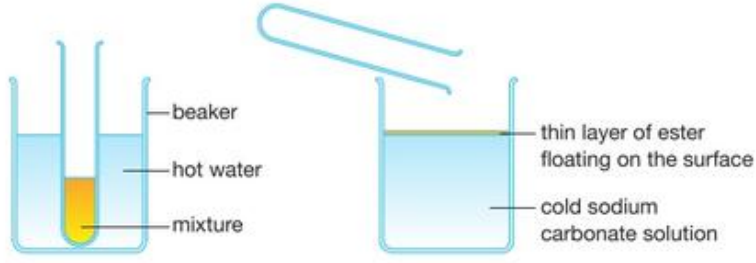
Safety Note: Wear goggles and dispense the acids in a fume cupboard. Avoid all skin contact with the acids; wash off immediately with plenty of cold water if you get some on your skin. Carefully follow the instructions as to how to smell the esters produced.



▲ Figure 28.5 Cosmetics researcher smelling a perfume under development.

It is easy to produce esters in the lab on a test-tube scale by reacting various combinations of carboxylic acids and alcohols in the presence of a few drops of concentrated sulfuric acid. The mixture has to be heated, but not by a direct Bunsen flame because the contents of the tubes are flammable. The following procedure could be used:

- Put 1 cm^3 of ethanoic acid and 1 cm^3 of ethanol into a boiling tube.
- Mix well and carefully add a few drops of concentrated sulfuric acid.
- Place the boiling tube in a beaker of hot water at about 80 °C for 5 minutes.
- Allow the contents of the tube to cool.
- When cool, pour the mixture into a beaker half-full of 0.5 mol/dm^3 sodium carbonate solution. There will be some effervescence as the excess acid reacts with the metal carbonate. The reason for doing this is to be able to smell the ester better. Not much ester is likely to be formed in such a short time, and the smell of the ester in the tube is going to be masked by the smell of the carboxylic acid.
- A layer of ester will separate and float on top of the water. The acid and alcohol will mostly dissolve in the water, but the ester won't. The small esters that you are likely to make in the lab float on the water, which makes them easy to smell.
- Smell the product by gently wafting the odour towards your nose with your hand. Do not put your nose near the top of the beaker!
- Repeat this procedure for other combinations of acids and alcohols. For example, you can make butyl butanoate, which should smell like pineapple, and propyl ethanoate, which smells like pear.



▲ Figure 28.6 Making esters on a test-tube scale. The ester is likely to be colourless but colour has been used in the diagram to make it easier to see.

EXTENSION WORK

Esters are hardly soluble in water whereas small alcohols and carboxylic acids usually are. This is because the –OH and the –COOH functional groups in alcohols and carboxylic acids can form a special type of intermolecular force called *hydrogen bonds* with water molecules. The attractions between water and alcohol/acid molecules are strong enough to dissolve these molecules. The lack of these functional groups (–OH and –COOH) in esters means that they form very weak interactions with water molecules and therefore will not dissolve.

CHAPTER QUESTIONS

SKILLS INTERPRETATION 7

1 a Draw the structural formulae for the following compounds:

i ethyl ethanoate

8 ii butyl butanoate

7 b Draw the displayed formulae for the following compounds:

i methyl butanoate ii ethyl propanoate

SKILLS CRITICAL THINKING

c Name the following compounds:

i $CH_3CH_2COOCH_3$ ii $HCOOCH_2CH_3$

SKILLS INTERPRETATION 8

2 Suppose you have access to bottles of ethanol, propan-1-ol, ethanoic acid, propanoic acid and concentrated sulfuric acid.

a Draw the structural formulae of all the esters you could make from combinations of the alcohols and carboxylic acids.

SKILLS EXECUTIVE FUNCTION 5

b Describe how you would make small samples of each of these esters.

6 c Explain how you could detect the formation of the esters in these reactions.

SKILLS CRITICAL THINKING 5

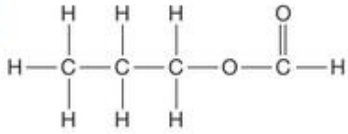
d Give two common uses for esters.

SKILLS CRITICAL THINKING, INTERPRETATION 8

3 Give the names and draw the structural formulae of the alcohols and carboxylic acids that could be used to make the following esters:

a methyl propanoate

b



c butyl ethanoate

SKILLS INTERPRETATION 7

4 Ethyl ethanoate can be made from ethanol as the only organic starting material in a number of steps.

a Draw the displayed formula for ethyl ethanoate.

SKILLS CRITICAL THINKING 5

b Identify the carboxylic acid that is required for the esterification of ethanol to form ethyl ethanoate.

c Describe how you could make the carboxylic acid in b from ethanol.

SKILLS PROBLEM SOLVING 9

d Write a balanced chemical equation for the formation of ethyl ethanoate from ethanol and the carboxylic acid in b.

SKILLS CRITICAL THINKING 5

e Write down the conditions required for the esterification step.

SKILLS EXECUTIVE FUNCTION 6

5 If you were given three unlabelled samples of ethanol, ethanoic acid and ethyl ethanoate, but weren't allowed to smell them, suggest how you might identify which one was which, with as much certainty as possible.

END OF CHEMISTRY ONLY